# HEARTLAND PROTOCOL
# Heart failure Evidence-based Access in Rural Treatment, Linking Advanced Network Delivery
# A Clinical Implementation Toolkit for Heart Failure Management in Resource-Limited and Underserved Settings

MASTER PROTOCOL — Version 3.3 — March 2026

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## Contents

# Executive Summary

Heart failure (HF) affects over 6.7 million Americans and causes more than 1 million hospitalizations annually. While urban centers benefit from specialist access, rural populations face a 53% higher mortality rate, exacerbated by limited cardiology access—86% of rural counties lack any cardiologist. This disparity is widening: young adults in rural areas experienced a 21% increase in cardiovascular mortality from 2010-2022, compared to only 3% in urban areas.

The **HEARTLAND Protocol** (Heart failure Evidence-based Access in Rural Treatment, Linking Advanced Network Delivery) addresses this crisis through a tiered implementation model designed for varying resource levels.

**Version 3.3 Strategic Features:**

1. **Tiered Implementation:** Three levels (Minimal/Standard/Advanced) accommodating diverse resource settings
2. **Dual-Track Execution:** Functions via Digital OR Analog pathways to bridge connectivity gaps
3. **2025 Pharmacology Integration:** Incorporates emerging evidence (finerenone, SGLT2i across phenotypes) with appropriate caveats regarding evolving guidelines
4. **Pharmacoeconomic Navigation:** Ensures no patient goes untreated due to cost through "Generic Bridge" safety net
5. **Task-Shifting Architecture:** Clear role delineation with alternatives when CHWs unavailable
6. **Value-Based Care Readiness:** Aligned with anticipated CMS payment model requirements
7. **Rural Risk Assessment Gap Analysis:** Evidence-based rationale for supplementary risk score incorporating distance-to-care and social support variables omitted by MAGGIC, GWTG-HF, and SHFM

**Core Philosophy:**

> *"HEARTLAND is designed to function across resource levels—from settings with advanced technology and specialist support to those relying on telephone calls and paper diaries. The goal is ensuring that resource limitations do not become barriers to evidence-based care."*

# Abbreviations

| Abbreviation | Definition |
|---|---|
| ARNI | Angiotensin Receptor-Neprilysin Inhibitor |
| CAH | Critical Access Hospital |
| CHW | Community Health Worker |

| Abbreviation | Definition |
|---|---|
| CKM | Cardiovascular-Kidney-Metabolic (AHA syndrome framework) |
| CRT | Cardiac Resyn-chronization Therapy |
| FQHC | Federally Qualified Health Center |
| GDMT | Guideline-Directed Medical Therapy |
| GLP-1 RA | Glucagon-Like Peptide-1 Receptor Agonist |
| HFmrEF | Heart Failure with Mildly Reduced Ejection Fraction (LVEF 41–49%) |
| HFpEF | Heart Failure with Preserved Ejection Fraction (LVEF ≥50%) |
| HFrEF | Heart Failure with Reduced Ejection Fraction (LVEF ≤40%) |
| HRRP | Hospital Readmissions Reduction Program |
| IVR | Interactive Voice Response |
| LVEF | Left Ventricular Ejection Fraction |

| Abbreviation | Definition |
|---|---|
| MRA | Mineralocorticoid Receptor Antagonist |
| NYHA | New York Heart Association (functional classification) |
| PAP | Patient Assistance Program |
| RHC | Rural Health Clinic |
| RPM | Remote Patient Monitoring |
| SBAR | Situation-Background-Assessment-Recommendation (handoff framework) |
| SGLT2i | Sodium-Glucose Cotransporter 2 Inhibitor |
| SMA | Shared Medical Appointment |

# Epidemiological Context

| Statistic | Impact | Source |
|---|---|---|
| 6.7 million Americans living with heart failure | Substantial disease burden | AHA 2025 |
| 86% of rural counties lack any cardiologist | Critical access gap | JACC 2024 |
| 53% higher cardiovascular mortality in rural vs. urban areas | Widening disparity | CDC/JACC 2024 |

| Statistic | Impact | Source |
|---|---|---|
| 21% increase in young adult rural CV mortality (2010-2022) | Concerning trend reversal | AJPM 2024 |
| $858 billion projected annual HF costs by 2050 | Unsustainable trajectory | AHA Advisory 2024 |
| CMS ASM 2027 value-based payment model announced | Financial implications for hospitals | CMS Innovation Center |

## Protocol Objectives

1. Standardize risk stratification using the AHA Cardiovascular-Kidney-Metabolic (CKM) framework
2. Optimize guideline-directed medical therapy (GDMT) through systematic titration protocols
3. Incorporate emerging 2025 evidence with appropriate clinical context
4. Structure discharge transitions using evidence-based teach-back methodology
5. Enable remote monitoring with dual digital/analog pathways
6. Provide tiered implementation bundles (Minimal/Standard/Advanced) for varying resource levels
7. Address non-pharmacological management comprehensively
8. Prepare facilities for evolving value-based payment requirements

## Methodological Approach

This protocol was developed through a targeted narrative review of high-impact clinical trials, meta-analyses, and authoritative guidelines (2018–2025), with focus on interventions validated in underserved populations or demonstrating feasibility for non-specialist delivery. It is an implementation framework, not a systematic review.

The protocol structure is informed by the Consolidated Framework for Implementation Research (CFIR; Damschroder et al., *Implement Sci* 2022), which provides a systematic approach to identifying factors that influence implementation success across diverse settings. CFIR domains—particularly intervention characteristics, outer setting, inner setting, and implementation process—guided the design of tiered implementation bundles and the dual-track execution system.

The global burden of heart failure—estimated at 64 million people worldwide (Savarese et al., *Cardiovasc Res* 2023)—underscores the potential international applicability of tiered implementation frameworks like HEARTLAND, although this protocol focuses specifically on rural and resource-limited settings in the United States.

## Target Settings and Implementation Tiers

### Setting Characteristics

| Setting Type | Characteristics | Recommended Tier |
|---|---|---|
| Critical Access Hospitals (CAHs) | <25 beds, limited staff, rural | Tier 1 (Minimal) or Tier 2 |
| Federally Qualified Health Centers (FQHCs) | 340B access, underserved populations | Tier 2 (Standard) |
| Rural Health Clinics (RHCs) | Limited specialist access | Tier 1 or Tier 2 |
| Indian Health Service Facilities | Tribal communities, unique barriers | Tier 1 with cultural adaptation |
| Primary Care Practices | Managing HF without cardiologist | Tier 2 |
| Regional Referral Centers | Hub for smaller facilities | Tier 3 (Advanced) |

## Implementation Tiers

The complete implementation tier specifications are summarized in Table 2.

Table 4: Table 2. Implementation Tiers Summary

| Component | Tier 1 (Minimal) | Tier 2 (Standard) |
|---|---|---|
| Risk Stratification | Score at discharge | Full CKM + Score |
| Guideline-directed medical therapy (GDMT) | ≥2 classes, prioritize sodium-glucose cotransporter-2 inhibitor (SGLT2i) + beta-blocker (BB) | Target all classes in 14 days |
| Monitoring | Track B (Analog) | Dual-track (A/B) |
| Discharge Education | Condensed teach-back (3 domains) | Full teach-back (8 domains) |
| Follow-up | 48–72 hour call, 14-day visit | 48-hour call, 7-day visit, weekly ×4 |
| Staffing | Registered nurse/medical assistant (RN/MA) + physician (MD) | RN champion, MA, PharmD |
| Community health worker (CHW) | Alternative/Family | High-risk only |
| Financial | Generic Bridge | Patient assistance program (PAP) pursuit + Generi |

**Principle:** *Each tier represents a clinically meaningful level of care matched to available resources. A Tier 1 facility achieving consistent 48-hour phone calls and ≥2 GDMT classes is delivering evidence-based care within its resource constraints.*

## Expected Outcomes

**Important Note:** The following effect sizes are derived from source trials testing individual interventions. The integrated HEARTLAND bundle has not been prospectively validated. Actual outcomes will depend on implementation fidelity, patient population, and local context.

| Outcome Measure | Effect Size from Source Trials | Evidence Source |
|---|---|---|
| 30-day readmission reduction | 30-45% (teach-back, transitional care) | Meta-analyses |
| All-cause mortality reduction | 19-35% (remote monitoring, GDMT) | TIM-HF2, GDMT trials |
| GDMT class addition | 53% absolute increase (telephone-based) | Hozhó Trial |
| HF events reduction (HFpEF) | 16% (finerenone in trial population) | FINEARTS-HF |
| Healthcare cost reduction | ~50% with optimized GDMT | STRONG-HF economic analysis |

The eight modules that follow translate these evidence-based targets into operational clinical workflows, progressing from initial risk assessment through pharmacotherapy, care transitions, monitoring, and quality measurement.

# Module 1: Risk Stratification

## 1.1 Cardiovascular-Kidney-Metabolic (CKM) Syndrome Framework

The AHA's 2023 Presidential Advisory established CKM syndrome as an integrated approach to cardiovascular risk—moving beyond isolated organ assessment. An estimated 89 million Americans are at risk across CKM stages.

| Stage | Description | Prevalence | Life Expectancy Impact |
|---|---|---|---|
| 0 | No CKM risk factors | 13.2% | Reference |
| 1 | Excess adiposity (BMI ≥25) and/or prediabetes | 20.8% | Minimal |

| Stage | Description | Prevalence | Life Expectancy Impact |
|---|---|---|---|
| 2 | Metabolic risk factors (T2DM, HTN, high TG) and/or moderate-high CKD | 53.1% | Moderate |
| 3 | Subclinical CVD with CKM | 5.0% | Significant |
| **4** | **Clinical CVD (including HF) ± kidney failure** | **7.8%** | **-12.4 years at age 50** |

**Clinical Relevance:** CKM staging helps identify patients requiring intensified multi-system management beyond traditional HF focus.

## 1.2 HEARTLAND Risk Score

**Rationale: Why Existing Risk Scores Are Insufficient for Rural Settings**

A comparison of existing risk scores with the HEARTLAND Risk Score is provided in Table 6.

Table 7: Table 6. Comparison of Heart Failure Risk Score Characteristics

| Characteristic | MAGGIC | GWTG-HF | SHFM | HEARTLAND |
|---|---|---|---|---|
| Number of variables | 13 | 7 | 24+ | **10** |
| Outcome predicted | 1–3 year mortality | In-hospital mortality | 1–5 year survival | **Readmission risk + monitoring intensity** |
| Distance to care | No | No | No | **Yes** |
| Social support | No | No | No | **Yes** |
| Rurality | No | No | No | **Yes** |
| Primary care feasibility | Moderate | Low (hospital-designed) | Low (complex) | **High** |
| Validation status | Validated (39,372 patients, 30 studies) | Validated (hospital registry) | Validated (multiple cohorts) | **Pragmatic heuristic (not yet validated)** |
| Intended use | Prognostic | Prognostic | Prognostic | **Clinical decision support** |

MAGGIC = Meta-Analysis Global Group in Chronic Heart Failure; GWTG-HF = Get With The Guidelines-Heart Failure; SHFM = Seattle Heart Failure Model; HEARTLAND = Heart failure

**Key evidence supporting rural-specific variables:**

- **Rurality and HF risk:** Population-based study demonstrated rurality independently associated with significantly higher risk of incident HF (Turecamo et al., JAMA Cardiol 2023)
- **Distance to care:** Mean distance to cardiology care reaches 87 miles in counties without a cardiologist vs. 16 miles with one (Kim et al., JACC 2024)

- **Social isolation:** Perceived social isolation associated with 3.74-fold increase in HF mortality (HR 3.74, 95% CI 1.82-7.70; Manemann et al., JAHA 2018)
- **Social deprivation:** Social deprivation indices significantly predict HF readmission independent of clinical severity (Deo et al., Circ Cardiovasc Qual Outcomes 2024)

**Important Note:** This score is a pragmatic heuristic designed to supplement—not replace—validated prognostic instruments such as the MAGGIC score. It has not been statistically validated through derivation/validation cohorts with ROC analysis or calibration testing. Formal validation using registry data linked with geographic and social determinant variables represents a planned next step in this research program.

The complete scoring reference is provided in Table 1.

Table 8: Table 1. HEARTLAND Risk Score Reference

| Risk Factor | Points |
|---|---|
| Age ≥75 years | +2 |
| Prior heart failure (HF) hospitalization within 6 months | +3 |
| Estimated glomerular filtration rate (eGFR) <45 mL/min/1.73m² | +2 |
| B-type natriuretic peptide (BNP) ≥500 pg/mL or N-terminal pro-B-type natriuretic peptide (NT-proBNP) ≥1500 pg/mL | +2 |
| Systolic blood pressure (BP) <100 mmHg at admission | +2 |
| Diabetes mellitus | +1 |
| Left ventricular ejection fraction (LVEF) <30% | +2 |
| Cardiovascular-kidney-metabolic (CKM) Stage 3 or 4 | +2 |
| Distance to cardiology care >50 miles | +1 |
| Lives alone or limited social support | +1 |

*Maximum Score: 18 points. LOW (0–4): Standard; MODERATE (5–8): Enhanced Bundle; HIGH (≥9): Intensive Bundle*

## 1.3 Risk-Stratified Care Pathways

| Score | Risk Level | Recommended Actions |
|---|---|---|
| 0-4 | LOW | Standard discharge; PCP follow-up within 14 days; Basic self-monitoring |
| 5-8 | MODERATE | Enhanced bundle; PCP follow-up within 7 days; 72h telehealth check; Home monitoring |

| Score | Risk Level | Recommended Actions |
|---|---|---|
| ≥9 | HIGH | Intensive bundle; Cardiology input before discharge; 48h phone follow-up; CHW or equivalent support |

With patients risk-stratified and care pathways assigned, the next step is defining the pharmacotherapy targets that will guide treatment optimization across all risk levels.

# Module 2: GDMT Optimization & 2025 Pharmacology

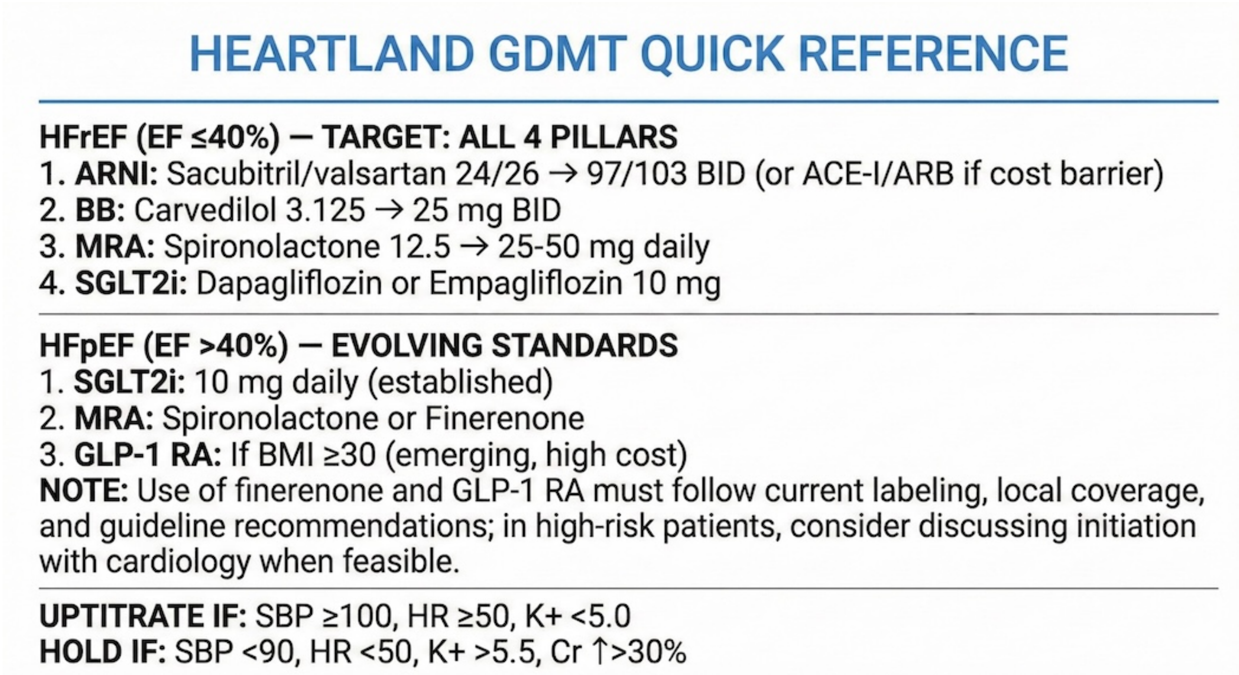
**Philosophy:** *"Some therapy is better than no therapy."* The protocol balances evidence-based medicine with practical constraints, ensuring no patient goes untreated while optimal therapy is pursued.

## 2.1 HFrEF (LVEF <=40%) — Quadruple Therapy

**Target:** Initiate all tolerated GDMT classes, with goal of achieving target doses through structured titration.

| Drug Class | Agent (Generic) | Starting Dose | Target Dose | Safety Gates |
|---|---|---|---|---|
| ARNI | Sacubitril/valsartan | 24/26 mg BID | 97/103 mg BID | SBP >100; K+ <5.5 |
| Beta-blocker | Carvedilol | 3.125 mg BID | 25 mg BID (50 if >85kg) | HR >50; SBP >90 |
| MRA | Spironolactone | 12.5-25 mg daily | 25-50 mg daily | eGFR >30; K+ <5.0 |
| SGLT2i | Dapagliflozin or Empagliflozin | 10 mg daily | 10 mg daily (no titration) | eGFR >20 |

A visual GDMT quick reference card is provided (Figure 1).


# HEARTLAND GDMT QUICK REFERENCE

**HFrEF (EF ≤40%) — TARGET: ALL 4 PILLARS**
1. **ARNI:** Sacubitril/valsartan 24/26 → 97/103 BID (or ACE-I/ARB if cost barrier)
2. **BB:** Carvedilol 3.125 → 25 mg BID
3. **MRA:** Spironolactone 12.5 → 25-50 mg daily
4. **SGLT2i:** Dapagliflozin or Empagliflozin 10 mg

**HFpEF (EF >40%) — EVOLVING STANDARDS**
1. **SGLT2i:** 10 mg daily (established)
2. **MRA:** Spironolactone or Finerenone
3. **GLP-1 RA:** If BMI ≥30 (emerging, high cost)

**NOTE:** Use of finerenone and GLP-1 RA must follow current labeling, local coverage, and guideline recommendations; in high-risk patients, consider discussing initiation with cardiology when feasible.

**UPTITRATE IF:** SBP ≥100, HR ≥50, K+ <5.0
**HOLD IF:** SBP <90, HR <50, K+ >5.5, Cr ↑>30%

Figure 1: Figure 1. Guideline-Directed Medical Therapy (GDMT) Quick Reference Card

## 2.2 HFmrEF / HFpEF (LVEF >40%) — Evolving Evidence

**Important Context:** The therapeutic landscape for HFpEF has evolved significantly. The EMPEROR-Preserved trial (Anker et al., *NEJM* 2021) demonstrated that empagliflozin reduces the combined risk of cardiovascular death or hospitalization for HF in patients with LVEF >40%, and the DELIVER trial (Solomon et al., *NEJM* 2022) confirmed these benefits with dapagliflozin across a broad HFpEF population, supporting a Class IIa recommendation for SGLT2i in HFpEF per the 2022 AHA/ACC/HFSA guideline. Additional recommendations below incorporate emerging evidence, and clinicians should be aware that guidelines continue to be updated.

| Priority | Agent (Generic) | Dose | Evidence Context |
|---|---|---|---|
| 1. SGLT2i | Dapagliflozin or Empagliflozin | 10 mg daily | Class IIa recommendation per 2022 AHA/ACC/HFSA guideline, supported by EMPEROR-Preserved and DELIVER trials. Reduces CV death & HF hospitalization regardless of diabetes status |

| Priority | Agent (Generic) | Dose | Evidence Context |
|---|---|---|---|
| 2. MRA | Finerenone OR Spironolactone | 10-20 mg / 12.5-25 mg | Finerenone: FINEARTS-HF showed 16% reduction in CV death/HF events (requires baseline K+ <5.0 and eGFR ≥25). Spironolactone: Established, low-cost option |
| 3. GLP-1 RA | Semaglutide | Titrate to 2.4 mg weekly | STEP-HFpEF: Improved symptoms in obesity phenotype (BMI ≥30). Best considered obesity therapy with CV benefits |
| 4. Diuretics | Loop diuretics | PRN | Symptom/volume control |

**Finerenone vs. Spironolactone Clinical Decision Guide**

**Safety Gates for Finerenone Initiation:** Baseline potassium <5.0 mEq/L and eGFR $≥$25 mL/min/1.73m² (per FINEARTS-HF inclusion criteria). Recheck potassium and eGFR within 1 week of initiation and after each dose change. Clinicians should be aware that guideline updates incorporating FINEARTS-HF data are ongoing.

| Clinical Scenario | Suggested Approach | Rationale |
|---|---|---|
| HFpEF with eGFR 25-60, K+ <5.0 | Consider finerenone | FINEARTS-HF population; requires eGFR ≥25 and K+ <5.0 |
| History of hyperkalemia on MRA | Finerenone preferred (monitor K+ closely) | Lower hyperkalemia incidence in trials; still requires K+ monitoring |
| Significant cost barrier | Spironolactone | ~$4/month generic vs. ~$500/month |
| HFrEF | Either acceptable | Both have supporting evidence |

| Clinical Scenario | Suggested Approach | Rationale |
|---|---|---|
| Uncertain, guideline-adherent approach | Spironolactone | Established guideline recommendation |

## 2.3 Non-Pharmacological Management

### Dietary Sodium Restriction

| Recommendatio | Details |
|---|---|
| Target | <2,000 mg/day (some guidelines suggest <1,500 mg for symptom control) |
| High-sodium foods to avoid | Processed meats, canned soups/vegetables, restaurant food, chips, cheese, bread |
| Practical tips | Cook at home; use herbs/spices instead of salt; read labels; rinse canned foods |
| Patient education | Provide written list of high/low sodium foods; teach label reading |

### Physical Activity and Exercise

| Recommendatio | Details |
|---|---|
| General guidance | Regular physical activity is recommended for stable HF patients |
| Starting point | Walking 5-10 minutes daily, gradually increase |

| Recommendatio | Details |
|---|---|
| Target | 30 minutes of moderate activity most days |
| Precautions | Rest when short of breath; avoid extreme temperatures; stop if chest pain/dizziness |
| Cardiac rehabilitation | Class I recommendation — refer all eligible patients |

## 2.4 The "Generic Bridge" & Pharmacoeconomic Navigation

**Problem:** Optimal agents (sacubitril/valsartan, finerenone, SGLT2 inhibitors, semaglutide) carry costs of $400-1,000+/month, prohibitive for uninsured/underinsured patients.

**Principle:** Leaving patients untreated while pursuing optimal agents is unacceptable. Generic therapy provides substantial benefit and is always preferable to no therapy.

**Generic Bridge Regimen (~$15/month total):**

1. ACE inhibitor (Lisinopril) OR ARB (Losartan) — $4/month
2. Beta-blocker (Carvedilol generic) — $4/month
3. MRA (Spironolactone generic) — $4/month
4. Metformin (if diabetic/prediabetic) — $4/month

**KEY PRINCIPLE:** Generic therapy is superior to NO therapy. Never delay treatment while waiting for paperwork.

The financial navigation tracker (Table 5) provides a structured tool for monitoring therapy optimization and PAP applications.

Table 15: Table 5. HEARTLAND Financial Navigation Tracker

| Field | Entry |
|---|---|
| Patient | [Text field] |
| Date | [Text field] |
| Insurance Type | Commercial / Medicare / Medicaid / Uninsured / 340B Eligible |
| Current Therapy | Generic Bridge / Partial Optimal / Full Optimal |

Patient Assistance Program (PAP) Applications: Track medications (ARNI, SGLT2i, Finerenone, GLP-1 RA) with Date Applied, Status (P = Pending, A = Approved, D = Denied), and Approva

While Module 2 defines the target regimens and dosing goals, achieving therapeutic doses in practice requires a structured titration process—particularly in settings where in-person visits are infrequent or logistically challenging.

# Module 3: Telephone-Based GDMT Titration

## 3.1 Evidence Base: The Hozho Trial

The Hozho Trial (JAMA Internal Medicine 2024) provides validation for telephone-based GDMT optimization in underserved populations:

| Parameter | Result |
|---|---|
| Population | 103 American Indians with HF in rural Navajo Nation |
| Primary Outcome | 66.2% vs 13.1% GDMT class addition at 30 days |
| Absolute Increase | 53% |
| Telehealth Completion | 80.5% adherence to phone visits |
| Safety | No increase in adverse events (6.6% vs 5.0%, p=0.51) |
| Method | Voice telephone calls (not smartphone apps) |

**Critical Implication:** Low-technology approaches achieve meaningful outcomes. Digital infrastructure is not required for protocol success.

## 3.2 Dual-Track Execution System

### Track A: Digital Pathway

For patients with smartphone, reliable connectivity, technological comfort.

| Component | Implementation |
|---|---|
| Symptom tracking | App-based daily entry |
| Vital signs | Bluetooth-enabled devices with auto-upload |

| Component | Implementation |
|---|---|
| Communication | Video visits, secure messaging |
| Data transmission | Automated to monitoring dashboard |

**Track B: Analog/Voice Pathway**

For patients with limited technology, poor connectivity, preference for voice.

| Component | Implementation |
|---|---|
| Symptom tracking | Paper Diary (provided at discharge) |
| Vital signs | Standard devices; patient reads display |
| Communication | Voice telephone calls (scheduled) |
| Data transmission | Verbal report; staff enters manually |

Both tracks follow identical clinical decision algorithms. Track selection is based on patient capability and preference, not clinical need. The track assignment form (Table 4) guides the assessment.

Table 19: Table 4. HEARTLAND Track Assignment Form

| Assessment Item | Response |
|---|---|
| Patient Name | [Text field] |
| Medical Record Number (MRN) | [Text field] |
| Risk Score | [Text field] |
| Smartphone with reliable connectivity? | Yes / No |
| Comfortable using apps? | Yes / No |
| Reliable telephone access? | Yes / No |
| Track Assignment | Track A (Digital) / Track B (Analog) / Hybrid |
| Implementation Tier | Tier 1 / Tier 2 / Tier 3 |
| Equipment Provided | Blood pressure (BP) cuff / Scale / Paper diary |
| Signature and Date | [Text field] |

### 3.3 Titration Decision Algorithm

| Parameter | Action |
|---|---|
| SBP ≥100 mmHg AND asymptomatic | UPTITRATE to next dose level |
| SBP 90-99 mmHg AND asymptomatic | HOLD current dose; reassess in 1 week |
| SBP <90 mmHg OR symptomatic hypotension | REDUCE dose or hold; consider cardiology input |
| HR <50 (for beta-blockers) | Reduce dose; if symptomatic, hold |
| K+ 5.0-5.5 | Reduce MRA/finerenone dose; recheck in 1 week |
| K+ >5.5 | HOLD MRA/finerenone and ARNI; urgent recheck; dietary counseling |
| Cr increase >30% | HOLD ARNI/MRA; evaluate; cardiology consult |

Titration protocols ensure patients reach therapeutic doses, but the critical vulnerability period following hospital discharge requires structured transitions to prevent early readmission and ensure continuity of care.

# Module 4: Structured Discharge Transitions

Transitional care interventions have demonstrated significant reductions in HF readmissions across multiple studies (Van Spall et al., *Eur J Heart Fail* 2017). The discharge bundle integrates medication reconciliation and bedside delivery practices informed by the PILL-CVD trial (Kripalani et al., *Ann Intern Med* 2012), which demonstrated that pharmacist-led interventions reduce clinically important medication errors after hospital discharge.

### 4.1 Discharge Bundle Components

| Component | Timing | Tier 1 | Tier 2/3 |
|---|---|---|---|
| Case Management Assessment | Within 24h of admission | If available | Required |
| Teach-Back Education | Before discharge | 3 core domains | 8 domains |
| Medication Reconciliation | Before discharge | Simplified | Full with PharmD |
| Follow-Up Scheduled | Before leaving | 14-day PCP | 7-day visit confirmed |

## 4.2 Task-Shifting Framework

**Core Principle:** Clinical decisions remain with licensed personnel. Task shifting applies to data collection and education delivery, not clinical judgment.

| Task | Optimal Executor | Alternatives | No CHW Available |
|---|---|---|---|
| Teach-back education | RN | LPN (RN supervise), Pharmacist | RN with extended time; video education |
| Discharge checklist | RN | MA with script, LPN | RN or provider |
| Daily weight/BP calls | RN, Pharmacist | MA, CHW, Automated IVR | MA with script; IVR system |
| Symptom assessment | RN | LPN → RN review | RN via phone |
| Red-flag triage | RN/Provider | NO SUBSTITUTION | NO SUBSTITUTION |
| Home device setup | Home Health RN | CHW, Family caregiver | Family with phone instruction |
| Financial/PAP navigation | Social Worker | CHW, Financial navigator | SW via phone; self-service resources |
| Dietary education | Dietitian | RN, CHW (with script) | Written materials; phone dietitian |

## 4.3 Teach-Back Core Domains

**Tier 1 (Minimal) — 3 Core Domains:**

| Domain | Key Point | Teach-Back Question |
|---|---|---|
| Daily Weight | Weigh same time daily; call if +3 lbs/2 days or +5 lbs/week | ”When should you call about your weight?” |
| Medications | Take daily even when feeling well; never stop without calling | ”Why take your medicines every day?” |
| Warning Signs | SOB, swelling, waking breathless → call clinic | ”What symptoms should worry you?” |

A comprehensive teach-back checklist is provided (Figure 5).

**TEACH-BACK CHECKLIST**

**TIER 1: CORE DOMAINS (Required for All)**

☐ **DAILY WEIGHT:** Weigh same time daily; call if +3lbs/2 days or +5lbs/week. (Teach-back question: “When should you call about your weight?”)

☐ **MEDICATIONS:** Take daily even when well; never stop without calling. (Teach-back question:“Why take your medicines every day?”)

☐ **WARNING SIGNS:** SOB, swelling, waking breathless → call clinic. (Teach-back question: “What symptoms should worry you?”)

**TIER 2/3: ADDITIONAL DOMAINS (Add to Core)**

☐ **WHAT IS HF:** Basic understanding of condition.

☐ **SODIUM RESTRICTION:** Target <2,000 mg/day; avoid high-sodium foods.

☐ **FLUID MANAGEMENT:** (If applicable) Daily fluid limit and counting.

☐ **DETAILED ’WHEN TO CALL”:** Review Red Flags card.

☐ **ACTIVITY GUIDANCE:** Gradual increase, listen to body, cardiac rehab.

Patient Name: _______________ Educator Signature: _______________ Date: _________

Figure 2: Figure 5. Teach-Back Checklist

### 4.4 Post-Discharge Contact Protocol

| Timing | Purpose | Tier 1 | Tier 2/3 |
|---|---|---|---|
| 48-72 hours | Medication check, side effects, barriers | Phone call | Phone or video |

| Timing | Purpose | Tier 1 | Tier 2/3 |
|---|---|---|---|
| Day 7 | Vitals, GDMT tolerability, weight trend | Phone acceptable | In-person or video preferred |
| Weeks 2-4 | Titration, adherence | Weekly calls | Structured protocol |
| High-risk | Intensive support | As resources allow | CHW engagement, frequent contact |

Once patients transition home, sustained monitoring becomes essential to detect decompensation early—before it leads to emergency department visits or rehospitalization.

# Module 5: Remote Patient Monitoring

## 5.1 Evidence Base

**TIM-HF2 Trial (Lancet 2018):**

| Outcome | Result |
|---|---|
| All-cause mortality | HR 0.70 (95% CI 0.50-0.96) — 30% reduction |
| Days lost to hospitalization | 4.88% vs 6.64% |
| Key finding | Patients living farther from cardiologists benefit most |

## 5.2 Red Flag Alert Criteria

| Finding | Action Required |
|---|---|
| Weight gain ≥3 lbs in 2 days | Call clinic same day |
| Weight gain ≥5 lbs in 1 week | Urgent evaluation within 24h |
| SBP <90 mmHg with symptoms | Hold GDMT; call provider |

| **Finding** | **Action Required** |
|---|---|
| SpO2 <92% at rest (if baseline normal) | Urgent evaluation |
| New/worsening dyspnea at rest | Same-day evaluation |
| **Chest pain, syncope** | **EMERGENCY — Call 911** |

A red flags alert card is provided for patient reference (Figure 2). A patient daily monitoring diary supports structured self-tracking (Figure 3).

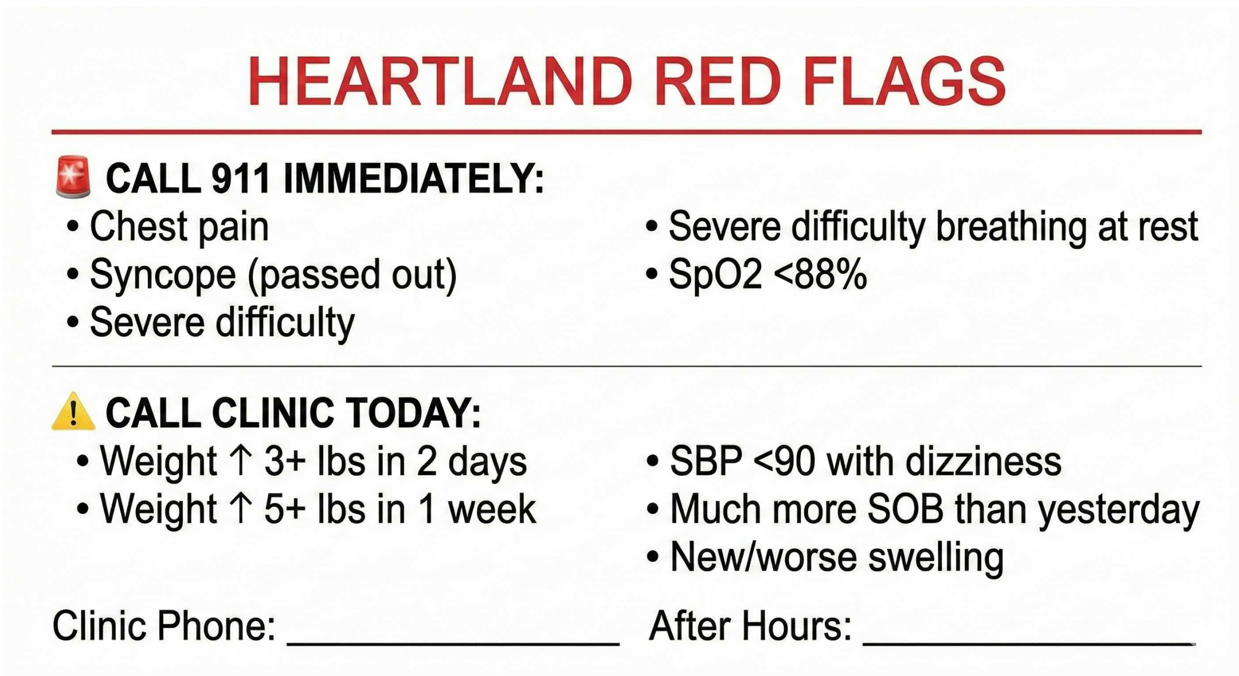
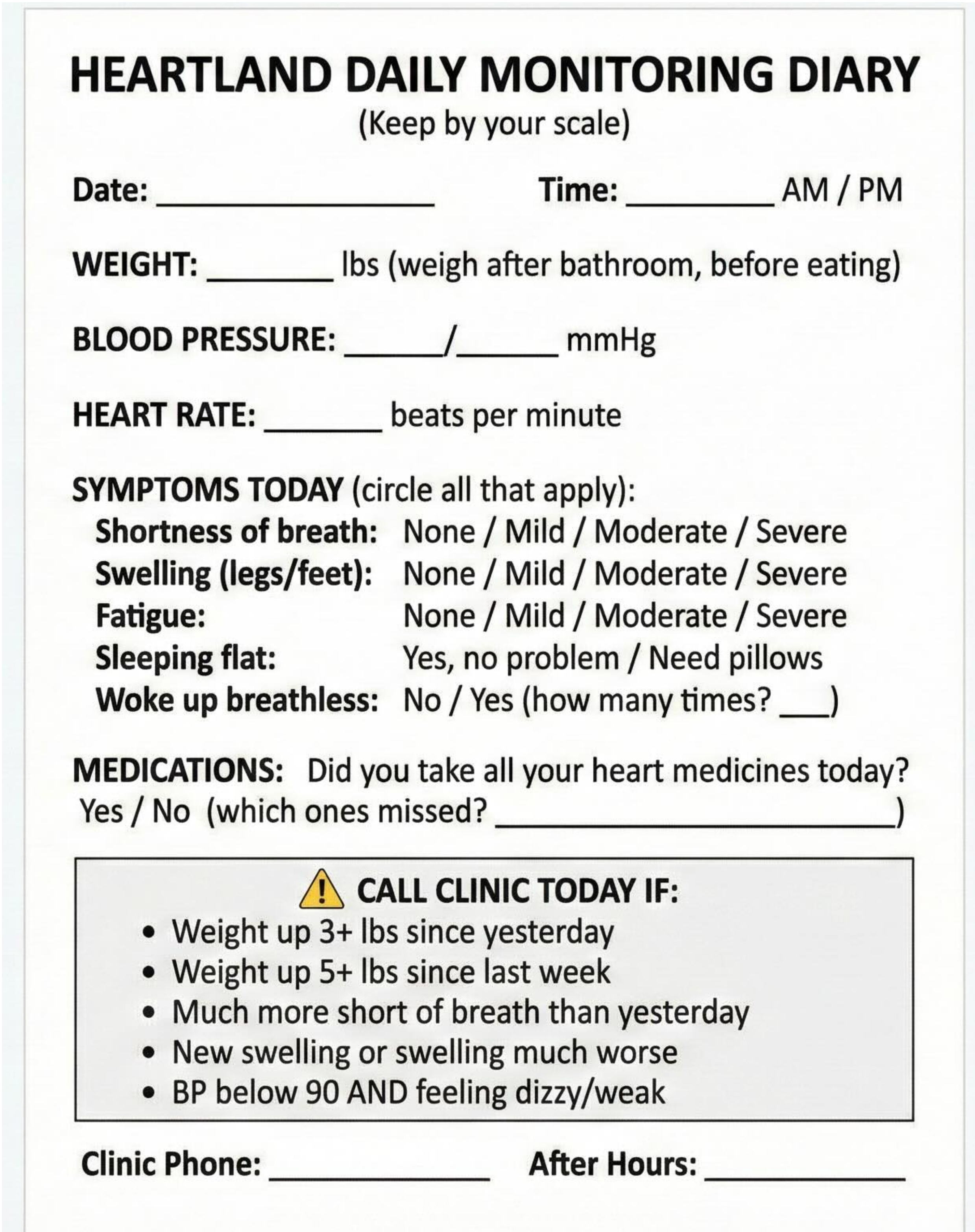



Figure 3: Figure 2. Red Flags Alert Card



# HEARTLAND DAILY MONITORING DIARY

(Keep by your scale)

**Date:** ____________________ **Time:** __________ AM / PM

**WEIGHT:** _________ lbs (weigh after bathroom, before eating)

**BLOOD PRESSURE:** ______/______ mmHg

**HEART RATE:** ________ beats per minute

**SYMPTOMS TODAY** (circle all that apply):

| | |
|---|---|
| **Shortness of breath:** | None / Mild / Moderate / Severe |
| **Swelling (legs/feet):** | None / Mild / Moderate / Severe |
| **Fatigue:** | None / Mild / Moderate / Severe |
| **Sleeping flat:** | Yes, no problem / Need pillows |
| **Woke up breathless:** | No / Yes (how many times? ___) |

**MEDICATIONS:** Did you take all your heart medicines today?
Yes / No (which ones missed? ___________________________)

**⚠ CALL CLINIC TODAY IF:**

- Weight up 3+ lbs since yesterday
- Weight up 5+ lbs since last week
- Much more short of breath than yesterday
- New swelling or swelling much worse
- BP below 90 AND feeling dizzy/weak

**Clinic Phone:** _______________ **After Hours:** ______________

Figure 4: Figure 3. Patient Daily Monitoring Diary

### 5.3 Financial Sustainability — Billing Codes (2025)

| Code | Description | Approximate Reimbursement |
|---|---|---|
| 99453 | RPM initial setup | $19-21 |
| 99454 | RPM monthly device (≥16 days data) | $48-55 |
| 99457 | RPM first 20 min management | $48-52 |
| 99458 | RPM additional 20 min | $38-42 |
| 98975-98981 | RTM codes (similar structure) | Similar range |
| G0511 | RHC/FQHC Comprehensive Care Management | Consolidated |

**Revenue Potential:** $150-200/month per high-risk patient with full capture.

Monitoring captures clinical changes, but heart failure rarely occurs in isolation—most patients carry multiple comorbidities that interact with HF management and require coordinated attention.

## Module 6: Comorbidity Management

### 6.1 Common Comorbidities — Practical Guidance

The comorbidity quick reference (Table 3) summarizes key considerations, recommended actions, and pitfalls to avoid.

Table 28: Table 3. Comorbidity Quick Reference

| Comorbidity | Key Considerations | What to Do |
|---|---|---|
| Atrial Fibrillation | Rate vs rhythm | Rate control, anticoagulation |
| Obstructive Sleep Apnea | Prevalent in heart failure with preserved ejection fraction (HFpEF) | Screen (STOP-BANG), continuous positive airway pressure (CPAP) |
| Iron Deficiency | Worsens symptoms | Check ferritin/transferrin saturation (TSAT), intravenous (IV) iron |
| Diabetes | Sodium-glucose cotransporter-2 inhibitor (SGLT2i) is dual therapy | SGLT2i, metformin, glucagon-like peptide-1 receptor agonist (GLP-1 RA |
| Chronic kidney disease (CKD) | Limits dosing, hyperkalemia | Adjust doses, monitor potassium, finerenone |
| Chronic obstructive pulmonary disease (COPD) | Beta-blocker (BB) concerns overstated | Cardioselective BB (metoprolol) |

Table 28: Table 3. Comorbidity Quick Reference

| Comorbidity | Key Considerations | What to Do |
|---|---|---|
| Depression | Affects adherence | Screen (Patient Health Questionnaire-9 [PHQ-9]), selective serotonin reup |
| Hypertension | GDMT treats both | GDMT lowers blood pressure (BP), adjust others |

### 6.2 Advanced HF Referral Criteria

**Refer for advanced therapies evaluation if:**

- LVEF <=35% despite >=3 months of optimal GDMT
- =2 HF hospitalizations in past 12 months
- Need for continuous or frequent IV inotropes
- Peak VO2 <14 mL/kg/min on CPET (if available)
- Persistent NYHA Class IIIb-IV symptoms
- Considering LVAD or transplant listing

**Refer for device evaluation (ICD/CRT) if:**

- LVEF <=35% after >=3 months of GDMT
- LBBB with QRS >=150 ms (CRT candidate)
- QRS 130-149 ms (possible CRT candidate)
- Primary prevention ICD consideration

**Special referrals:**

- Suspected cardiac amyloidosis (HFpEF + LVH + age >65 + carpal tunnel/neuropathy) — Needs PYP scan, cardiology
- Palliative care/hospice discussion when refractory symptoms, recurrent hospitalizations with declining trajectory, or patient/family goals shift toward comfort

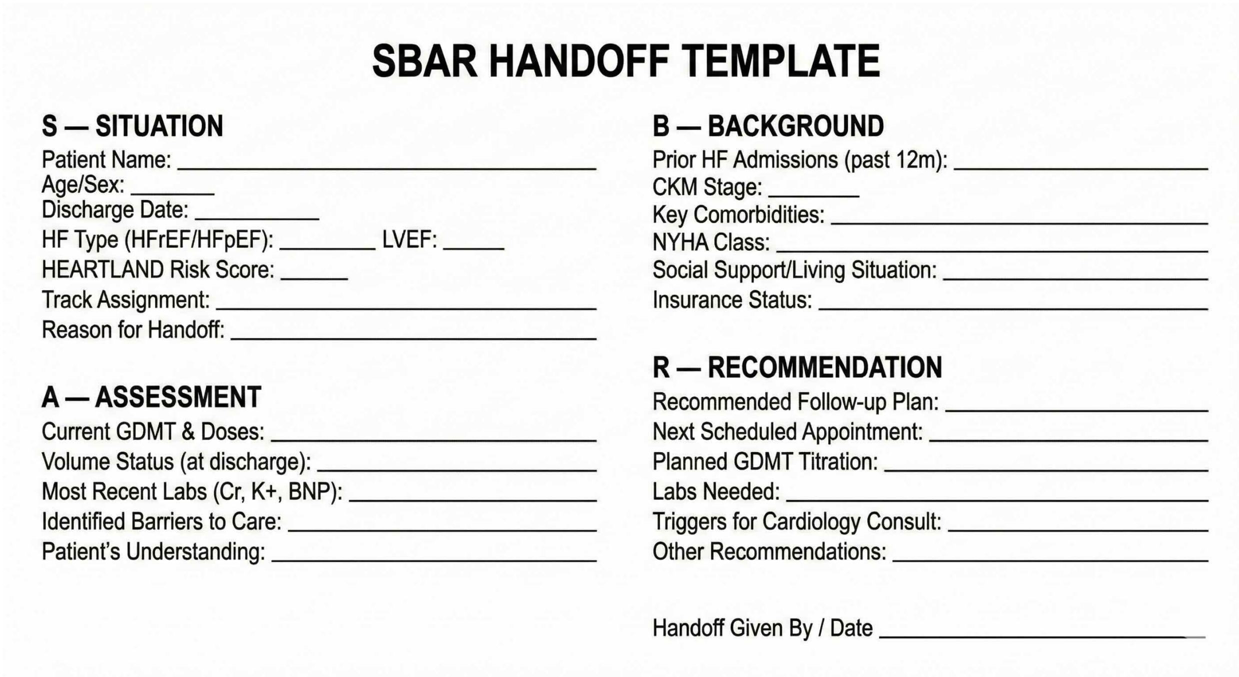
Effective comorbidity management requires coordination across providers and settings—particularly when patients receive care from multiple clinicians who may not share a common electronic health record.

## Module 7: Primary Care Coordination

### 7.1 SBAR Handoff Communication

| Element | Content for HF Handoff |
| --- | --- |
| S — Situation | Patient discharged with HF; Type (HFrEF/HFpEF); LVEF; Trigger; HEARTLAND Risk Score; Track Assignment |
| B — Background | Prior admissions; CKM stage; Key comorbidities; NYHA class; Social support; Insurance |
| A — Assessment | Current GDMT with doses; Volume status; Recent labs; Financial barriers |
| R — Recom-mendation | Follow-up interval; Titration plan; Labs to order; When to request cardiology input |

An SBAR handoff template is provided (Figure 4).



## SBAR HANDOFF TEMPLATE

**S — SITUATION**

Patient Name: ____________________
Age/Sex: ________
Discharge Date: ___________
HF Type (HFrEF/HFpEF): ________ LVEF: _____
HEARTLAND Risk Score: ______
Track Assignment: ____________________
Reason for Handoff: ____________________

**B — BACKGROUND**

Prior HF Admissions (past 12m): ____________________
CKM Stage: ________
Key Comorbidities: ____________________
NYHA Class: ____________________
Social Support/Living Situation: ____________________
Insurance Status: ____________________

**A — ASSESSMENT**

Current GDMT & Doses: ____________________
Volume Status (at discharge): ____________________
Most Recent Labs (Cr, K+, BNP): ____________________
Identified Barriers to Care: ____________________
Patient's Understanding: ____________________

**R — RECOMMENDATION**

Recommended Follow-up Plan: ____________________
Next Scheduled Appointment: ____________________
Planned GDMT Titration: ____________________
Labs Needed: ____________________
Triggers for Cardiology Consult: ____________________
Other Recommendations: ____________________

Handoff Given By / Date ____________________

Figure 5: Figure 4. SBAR Handoff Template

## 7.2 PCP Empowerment — What Can Be Managed Independently

Actions PCPs can confidently manage without specialist referral:

- GDMT titration per Module 3 algorithm when hemodynamically stable
- Diuretic adjustments for mild volume overload
- Finerenone initiation for HFpEF (if comfortable, with monitoring plan)
- Routine lab monitoring (BMP q1-2 weeks during titration, then quarterly)
- Vaccinations (influenza, pneumococcal, COVID-19)
- Depression/anxiety screening and treatment
- Cardiac rehabilitation referral
- Smoking cessation
- Diabetes co-management with HF-appropriate medications
- Iron deficiency screening and IV iron referral

## 7.3 Shared Medical Appointments (SMAs) — Tier 2/3

**Purpose:** Increase provider efficiency for stable HF patients.

| Component | Duration | Content |
|---|---|---|
| Group Education | 45-60 min | Sodium, weights, medications, warning signs |

| Component | Duration | Content |
|---|---|---|
| Individual Assessment | 5-10 min/patient | Vitals, symptom check, dose adjustments |

**Candidates:** NYHA I-II, stable GDMT >=4 weeks, willing to participate. Efficiency Gain: A single group session with 8–10 patients effectively replaces an equivalent number of individual encounters of similar duration, substantially increasing provider capacity for stable patient management.

Coordination structures are only as effective as the metrics used to evaluate them. Without systematic quality measurement, it is impossible to identify gaps, track improvement, or demonstrate value to payers and institutional leadership.

# Module 8: Implementation Guidance

## 8.1 Quality Metrics

| Metric | Tier 1 Target | Tier 2/3 Target |
|---|---|---|
| 48-72h post-discharge contact | >70% | >90% |
| ≥2 GDMT classes at discharge (HFrEF) | >60% | >80% |
| 7-day follow-up attendance | >60% | >85% |
| 30-day readmission rate | Improvement from baseline | <15% |
| Teach-back documentation | >70% | >90% |

**Principle:** Focus on incremental improvement. A facility moving from 30% to 50% contact rate is succeeding.

**Tier 3 Rapid Up-Titration:** For Tier 3 (Advanced) facilities, the STRONG-HF trial (Mebazaa et al., *Lancet* 2022) demonstrated that intensive, early optimization of HF therapies within two weeks of discharge reduces the composite of HF rehospitalization and all-cause death. Post-hospitalization initiation strategies (DeVore et al., *Circulation* 2020) further support this approach. This rapid-sequence model requires close monitoring and is appropriate only for settings with adequate staffing and infrastructure.

## 8.2 Value-Based Payment Preparation

**Context:** CMS has announced the Ambulatory Specialty Model (ASM) as a mandatory value-based payment program for heart failure, with implementation anticipated in 2027. Facilities should monitor CMS announcements for updates.

**Preparation Actions:**

- Register for CMS Innovation Center updates

- Establish baseline quality metrics
- Implement structured GDMT optimization
- Document telehealth capabilities
- Review HRRP readmission history
- Establish specialist teleconsult relationships

## Limitations and Scope

This protocol has several important limitations that users should consider when adapting it for local implementation:

1. **No prospective validation.** The HEARTLAND Protocol has not been tested in a prospective clinical trial or pilot implementation study. The expected outcomes reported in this document are extrapolated from source trials that tested individual interventions, not the integrated bundle.
2. **Risk score not statistically calibrated.** The HEARTLAND Risk Score is a pragmatic heuristic. It has not undergone derivation/validation cohort testing, ROC analysis, or calibration assessment. The variable weights reflect clinical reasoning informed by published evidence, not regression coefficients from a derivation dataset.
3. **Narrative review methodology.** The evidence synthesis underlying this protocol is based on a targeted narrative review, not a systematic review with predefined search strategies, inclusion/exclusion criteria, or risk-of-bias assessment.
4. **Clinical judgment remains paramount.** This protocol provides a structured framework but does not substitute for clinical judgment. Individual patient factors—including preferences, comorbidities, and social circumstances—should guide clinical decisions.
5. **Population-specific adaptations may be necessary.** The protocol was designed for rural and resource-limited US settings. Application to other populations (urban underserved, international settings, specific ethnic communities) may require modifications to account for local healthcare infrastructure, cultural factors, and regulatory environments.

## Appendix A: Evidence Base

### Key Clinical Trials

| Trial | Year | Key Finding | Context |
|---|---|---|---|
| EMPEROR-Preserved | 2021 | Empagliflozin reduces CV death/HF hospitalization in HFpEF (LVEF >40%) | Class IIa per 2022 AHA/ACC/HFSA guideline |
| DELIVER | 2022 | Dapagliflozin confirms SGLT2i benefit across broad HFpEF population | Class IIa per 2022 AHA/ACC/HFSA guideline |

| Trial | Year | Key Finding | Context |
|---|---|---|---|
| FINEARTS-HF | 2024 | Finerenone 16% reduction CV death/HF events in HFm-rEF/HFpEF | Emerging evidence; guideline integration ongoing |
| STEP-HFpEF | 2023 | Semaglutide improved symptoms in obese HFpEF | Obesity therapy with HF benefits |
| Hozhó Trial | 2024 | 53% increase GDMT via telephone in rural population | Validates low-tech approach |
| STRONG-HF | 2022 | Rapid GDMT titration safe and effective | Tier 3 methodology |
| TIM-HF2 | 2018 | 30% mortality reduction with remote monitoring | Foundation for RPM |

# Appendix B: Manufacturer Assistance Programs

*Note: Program details change frequently. Verify current availability before applying.*

| Medication | Manufacturer | Resource |
|---|---|---|
| Sacubitril/valsartan | Novartis | Patient assistance program available |
| Finerenone | Bayer | Savings program available |
| Empagliflozin | Lilly/BI | Patient assistance available |
| Dapagliflozin | AstraZeneca | Patient assistance available |
| Semaglutide | Novo Nordisk | Patient assistance available |

**Additional Resources:** NeedyMeds.org, RxAssist.org, 340B pricing for eligible facilities (FQHC, CAH, certain hospitals).

*HEARTLAND Protocol v3.3 | March 2026 | Vicky Muller Ferreira, MD*